

Adama Science and Technology University
School of Business and Economics
Department of Management
Statistics for Management-I

Individual Assignment- last submission date April 21, 2013, 9:00AM – 3:00PM

1. When drawing a card from a deck of playing card, what is the probability of drawing either an ace card or king in a single draw?
2. When drawing a card from a deck of playing card, what is the probability of drawing ace card or spade in a single draw?
3. Of 300 business students, 100 are currently enrolled in financial accounting and 80 are currently enrolled in Business statistics. These enrollment figures include 30 students who are in fact enrolled in both courses. What is the probability that randomly chosen student will be enrolled in either accounting or business statistics?
4. The following table shows annual house hold income for a number of families

Category	Income range (in birr)	Number of families
1	Less than 20,000	60
2	20,000 – 40,000	100
3	40,000 – 60,000	160
4	60,000 – 100,000	140
5	100,000 and above	40

With the reference to the above table, what is the probability of a randomly chosen family will have household income:

- a. Between birr 20,000 and birr 40,000?
 - b. Less than birr 40,000?
 - c. At one of the two extremes of being either less than birr 20,000 or at least birr 100,000?
5. The BOD of CBE consists of 8 men and 4 women. A four member committee is to be chosen at random to recommend a new branch president. What is the probability that:
 - A. All the members will be men?
 - B. All the members are women?
 - C. The first will be women and the fourth men?
 - D. The first men, second two women, and the last man?
 - E. The first man, second two women, and the fourth women?
 6. The following is contingency table which presents voter reactions to a new property tax plan according to party affiliation.

Party affiliation	Reaction		
	In favor	Neutral	Opposed
Democratic	120	20	20
Republican	50	30	60
Independent	50	10	40

- A. Prepare joint probability table and determine the marginal probabilities.

- B. What is the probability that the voter is a Republican given that the person is opposed to the plan?
- C. What is the probability that the voter is either a Democratic or a Republican?
- D. What is the probability that the voter is either a Democratic or in favor of the proposal?
7. The number of vans that have been requested for the rental at a car rental agency during last a given day period is identified in the following table.

Possible demand	Number of days
3	3
4	7
5	12
6	14
7	10
8	4

- A. What is the probability of each van being requested?
- B. Calculate the expected value and standard deviation for van requested.
8. A student doesn't know the answers to five multiple choices questions, each having four alternatives. She/he simply guesses the answers.
- A. Construct probability distribution for the number of correct responses.
- B. What is the probability of the student answering:
- Exactly 3 questions?
 - No questions correctly?
 - At least one questions correctly?
 - More than one questions correctly?
 - Less than three questions correctly?
 - At most two questions correctly?
- C. Calculate the mean and standard deviation of the distribution
9. Suppose 40% of the people entering a store make a purchase. If 10 people enter a store, find the expected number and standard deviation of people making a purchase.
10. Assume that patients arrive at the emergency room of ASTU Hospital at the average rate of 6 per hour on weekend evening.
- A. What is the probability of
- 4 arrivals in 30 minutes on a weekend evening?
 - At least 3 arrivals in 60 minutes on a weekend evening?
- B. Compute mean and standard deviation of the distribution.
11. If the average number of traffic accident in Adama city is 2 per week, what is the probability of
- No accident during the first month?
 - At most three accidents during a two weeks period?
12. The weekly income of large group of middle level managers are normally distributed with a mean of 1000 birr and standard deviation of birr 100.
- A. What is the Z-value for a manager who earns:
- Birr 1100?
 - Birr 900?
 - Birr 1256?
 - Birr 728?
- B. What is the probability of selecting a middle manager whose monthly income is:
- Between birr 1000 and birr 1100?

- ii. Less than birr 1100?
 - iii. Between birr 790 and birr 1000?
 - iv. Less than birr 790?
 - v. Between birr 840 and birr 1200?
 - vi. Between birr 1150 and birr 1250?
13. The age of a group of babies is normally distributed with mean is 3 years and standard deviation of 2 years. What is the probability that a randomly selected baby is:
- A. More than 4 years of age?
 - B. Between 2 and 5 years of age?
 - C. Less than 2 years of age?
 - D. Between 1 and 2 years of age?
 - E. Find the age below which 80% of babies have it?
 - F. Find the age above which 60% of babies have it?